Assignment 2 - Submission Requirements Audit

Team Plot Twist | 31 October 2026

This audit checks all three pages of DAS732_T1-26-27_A2.pdf against the supplied report, source code, data and runtime evidence. Technical coverage is documented below. Dataset registration and actual personal implementation require team confirmation. No assessment score is predicted.

Clause	Requirement	Status	Evidence and interpretation
Task / SciVis	Python color, contour and quiver implementations	Covered	Figs. 18-20, 27; scripts/build_scivis.py. Four real gridMET fields; no synthetic weather values.
Task / GIFs	Scalar animations with 5-10 time instances	Covered	Three supplied GIFs, each with eight frames: fixed-scale temperature, daily-scale temperature and contours. images/scivis/.
Task / InfoVis i	NetworkX or Gephi; at least three layouts and interactions	Covered	NetworkX Spring, Spectral and Circular; Fig. 1. web/network.html supports layout switching, threshold, country search, selection, pan and zoom.
Task / InfoVis ii-a	JavaScript parallel coordinates with edge bundling and interaction	Covered	Figs. 4-5; web/app.js and output/pcp_changes/app.js. Bundling, real pointer brushing, axis ordering and country inspection are tested. Bundles use explicit adoption groups, not inferred clusters.
Task / InfoVis ii-b	JavaScript zoomable treemaps	Covered	Figs. 9-10; web/treemap.js. Zoom/back, breadcrumbs and two complete hierarchy remodels retain the same 184 country leaves.
Task / InfoVis iii	Python horizon and ridgeline visualizations	Covered	Figs. 14-17; scripts/build_temporal_figures.py. Two horizon variations and two ridgeline variations.
Data / SciVis	gridMET and a contiguous three-month interval	Covered	July 1-September 30, 2023. The 92-day cache and separate eight-date cache retain the selected gridMET variables and acquisition provenance.
Data / rationale	Explain both the period and temporal sampling choice	Covered	Data and preparation: summer-to-autumn heat/cooling/rain timing; documented 2023 Southwest heat context; uniform 13-day spacing with endpoints, contrasted with random gaps. Full-period claims use all 92 dates.
Data / registration	Team slice selected through the TAs form; first-come policy	Verify	The supplied files cannot prove registration or uniqueness in the TAs form. Team must confirm its registered slice. This is an administrative check, not a plot feature.
Data / NetCDF	Extract vs and th for wind; other variables for scalars	Covered	vs maps to wind_speed and th to wind_from_direction in build_scivis.py; tmmx maps to air_temperature and pr to precipitation_amount. Source masks/packing are decoded; source Kelvin converts once to Celsius.
Data / A1	Build network, multivariate and hierarchy data from A1	Covered	Three preserved OWID/ITU extracts in data/raw/. Preprocessing outer-joins on entity/code/year, excludes 823 aggregate rows and retains 12,564 country-year rows without imputation.
Data / instructor	Discuss network difficulty with instructor before deadline	Covered	User-supplied instructor reply approves a derived similarity network and advises distribution-based filtering. Report attributes guidance once; P95 is explicitly the team choice. The email itself is not published.
Spec. 1 / bundle	Submit report, images folder and codebase	Covered	DAS732_A2_Report.pdf; images/report/Fig1.png-Fig32.png; data/, scripts/, web/, requirements.txt and launcher. Submission is self-contained without a Git remote.
Spec. 1a	Captioned report images and image-specific inferences	Covered	All 32 figure sets have a key-result caption, concise reading guide, evidence-based interpretation and numbered action takeaway. Full question/rationale guides are included in figures.html; common caveats are centralized.

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Spec. 1b	High-resolution assessment images with report indices	Covered	Numbered PNGs and available matching SVGs in images/report/. Raw plot exports also remain at their source paths for exact regeneration.
Spec. 1c	Titles, legends, axes and other necessary text	Covered	Rendered report and full-resolution exports reviewed. Units, thresholds, cohort/sample sizes and scale keys are included where relevant; non-Cartesian views use appropriate labels and legends.
Spec. 1d	Complete high-resolution GUI screenshots	Covered	The first-page dashboard link opens working browser UIs; images/gui/ and interactions.html retain complete high-resolution captures and extra states. The report PNGs are analytical chart exports.
Spec. 1e	README for implementation	Covered	Root README contains a local launch route, dependencies, data preprocessing, transformations, staged reproduction and validation commands. web/README.md explains interactions.
Spec. 1f	Video demo is not required	Not required	The supplied PDF explicitly says a video demo is not required. Old slides, scripts and unrecorded video materials are excluded from the core submission.
Spec. 1g	Professional report; appropriate references encouraged	Covered	IEEE-style report with complete figure explanations, readable captions and eleven cited references including gridMET, OWID, NetworkX, D3, Gephi and ContourPy/Marching Squares.
Spec. 2 / workload	Each student implements at least one SciVis and one InfoVis task	Verify	The assigned responsibilities satisfy the domain rule: Parikshith network/PCP + color; Sindhoor horizon/ridgeline + contours; Varshith treemap + quiver. Actual independent implementation is a team attestation, not verified by counts.
Spec. 2a	Combined outcome proportional to team strength	Covered	All specified visualization families and experiments are included for the three-person team, plus supporting analyses addressing the main question.
Spec. 2b	Report clearly states contributions	Covered	Named task-responsibility table; Person A/B/C and video duties removed. Figure allocation is 12/10/10. Assigned responsibilities must match actual independent work.
Spec. 2c-d	No joint task ownership; no two students do the same task	Verify	The assigned type ownership is distinct and all 32 figures have one assigned name. Shared preprocessing and reuse of method outputs are separated from task ownership. Confirm that actual work matches this table.
Spec. 3	Marching Squares and/or filled contours; explain rationale	Covered	Fig. 20; independent 16-case code, both saddle-center resolutions and missing-cell handling. All 32 case checks pass; real-grid segments match ContourPy at four levels. Lines/filled bands serve threshold/regional reading tasks.
Spec. 4	Continuous/discrete/logarithmic mappings and palettes	Covered	Fig. 18 compares continuous/discrete sequential, linear/log rain, a turbo baseline and a meaningful zero-centered departure map. Fig. 22 uses diverging differences. Quantity changes and zero-rain handling are explicit.
Spec. 5	Global versus local extrema across timesteps	Covered	Fig. 19 plus fixed/daily-scale GIFs; identical colors under different extrema can represent different temperatures. Global bounds cover the eight sampled fields, not every day in the interval.
Spec. 6a or 6b	Quiver sampling/length comparison OR streamline experiment	Covered	Option 6a: Fig. 27 compares 4/12 display-cell steps and speed-proportional versus same-length arrows. vs/th conversion, map bearing and 5 m/s key are explicit. Streamlines are optional and not implemented.
Spec. 7	Node-link layouts; node filtering optional	Covered	Three layouts in Fig. 1 and the live UI. Optional region and hide-isolate node filters are implemented in addition to required edge-threshold interaction.

Clause	Requirement	Status	Evidence and interpretation
Spec. 8	PCP must have axis reordering and/or brushing	Covered	Both are implemented and exercised through actual pointer/axis controls. Bundling changes paths while preserving measured values at the axes.
Spec. 9	Treemap hierarchy remodels; spatial partition optional	Covered	Region -> country and adoption band -> region -> country. Optional squarified versus slice-and-dice selector preserves totals and hierarchy. Figs. 9-10 plus UI evidence.
Spec. 10	At least two horizon and two ridgeline variations	Covered	Horizon: selected country series and regional medians (Figs. 14-15). Ridgeline: global yearly and regional yearly distributions (Figs. 16-17). Fixed cohorts and 0-100 scales avoid changing sample definitions.
Rubric / completeness	4 points: all experiments and complete deliverables	Evidence mapped	Every compulsory experiment maps to a figure, implementation or GIF above. Folder integrity and local runtime links are checked; external registration and actual ownership remain separate checks.
Rubric / correctness	8 points: valid inference for dataset space/time	Evidence mapped	Numerical verifiers recompute edges, distributions, paired changes, monthly weather, rainfall concentration and run lengths. Similarity is not a physical cable; observational associations are not causal claims.
Rubric / inferences	8 points: useful complex inferences from visualization	Evidence mapped	32 question/evidence/inference/takeaway sequences connect profiles, paired trajectories, within-group variation, full-month versus endpoint change, daily order and joint occurrence.
Rubric / data handling	10 points: preprocessing and transformations	Evidence mapped	Separate preprocess_data.py; raw extracts, region lookup, cache metadata and SHA-256 audit. Missingness/cohort rules, masks, temperature units, wind convention, subsampling and area weights are documented.

Before submission

Confirm dataset registration and actual independent task ownership. The responsibility statement must accurately reflect actual work. A video is not required by the supplied assignment PDF. The rubric totals 30 points (4 completeness, 8 correctness, 8 inferences, 10 data handling), scaled to 20.